

Bond Risk & Key Rate Duration Analysis

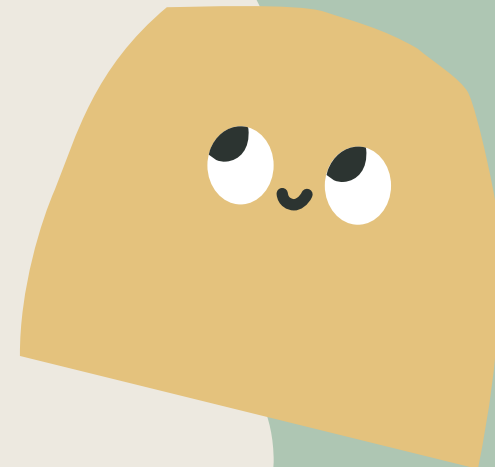
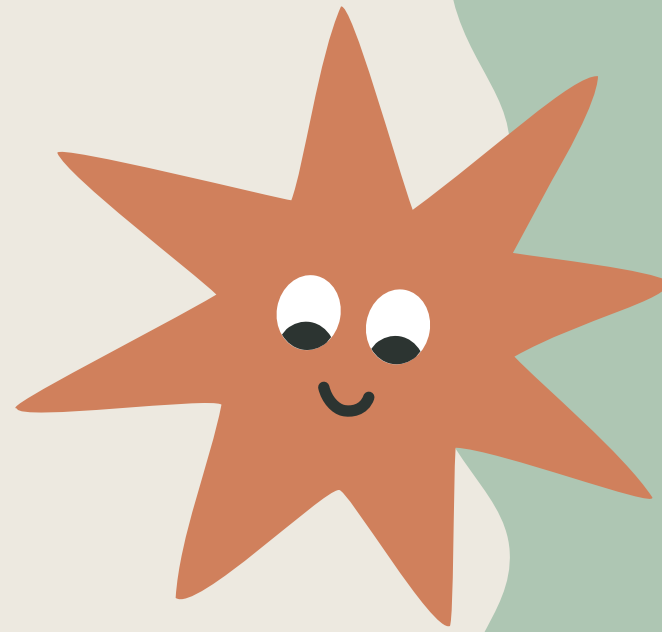
Internship Project Presentation

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1. Project Overview

- Analysis of bond interest-rate risk
- Focus on Key Rate Duration (KRD)
- Portfolio risk and DV01 analysis
- Yield-curve scenario analysis
- Machine learning-based risk analysis
- Visualization of analytical results

2. Project Objectives

- Analyse bond interest-rate sensitivity
- Calculate and interpret duration and convexity
- Measure Key Rate Duration across maturity buckets
- Identify major sources of portfolio risk
- Evaluate scenario-based P&L impact
- Apply machine learning for risk analysis
- Generate decision-supporting visualizations

3. Bond Mathematics & Risk Measures

- Bond Price
- Yield to Maturity
- Macaulay Duration
- Modified Duration
- Convexity
- DV01

These measures provide the foundation for fixed-income interest-rate risk analysis.

4. Key Rate Duration Analysis

KRD measures interest-rate sensitivity at specific maturity points of the yield curve.

Key maturity buckets:

3M | 6M | 1Y | 2Y | 3Y | 5Y | 7Y | 10Y | 20Y | 30Y

- Identifies concentration of interest-rate risk
- Captures non-parallel yield-curve movements
- Supports bucket-level risk interpretation

5. Portfolio Risk Analysis

- Portfolio Modified Duration
- Portfolio Convexity
- Total DV01
- Risk contribution by maturity bucket
- Risk concentration analysis
- Portfolio sensitivity to yield movements

6. Scenario Analysis

- Interest-rate shock scenarios
- Key-rate bucket P&L impact
- Yield-curve sensitivity
- Steepening scenario analysis
- Identification of high-impact maturity buckets

7. Machine Learning Analysis

- Predictive modelling applied to bond-risk data
- Model evaluation
- Feature importance analysis
- Actual vs predicted risk analysis
- ML used as a complementary risk-analysis approach

8. Results & Key Findings

- KRD provides a detailed view of yield-curve exposure
- Risk is distributed differently across maturity buckets
- Scenario analysis highlights potential P&L sensitivity
- DV01 helps quantify one-basis-point interest-rate exposure
- Feature importance identifies relevant predictive variables
- Results support more focused bond-risk monitoring

9. Conclusion & Future Scope

Conclusion:

- Bond risk was analysed using quantitative fixed-income measures
- KRD helped identify maturity-specific interest-rate exposure
- Scenario analysis provided portfolio impact insights
- Machine learning complemented traditional risk analysis

Future Scope:

- Real-time market data integration
- Advanced ML models
- Automated Power BI dashboards
- Expanded stress-testing scenarios
- Hedging and risk-optimization analysis

Thank You

Questions?